EL NIÑO ACTION BRIEFING · 13 OCTOBER 2026

El Progreso, Yoro, Honduras

In El Progreso, El Niño arrives as a dry season short of water. Short water means stored water, and stored water means dengue, which has killed the town's children in recent epidemics. But the first hazard is not dry at all. The Ulúa is already coming through levee breaches that have stood open since 2020, and the cold-front rains are still to come. Close the breaches, get the hospital ready for dengue, and keep every water container covered.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
The Climate Impact Lab region containing El Progreso, Yoro (municipality of about 200,000–230,000; regions hold about 300,000)	None published for Honduras, Yoro, or El Progreso	Jun 2026 – Feb 2027, heat only; the global six-month figure is 239,000 ± 11,000	Flooding through open Ulúa levee breaches now; then a dry-season water shortage (Dec–May) and the dengue that follows it (Mar–Oct 2027)

20/20 recommendations checked. Of 14 proposed actions, 13 were corrected (partially confirmed) and 1 is unverified (asking the state grain agency to sell discounted beans in El Progreso, where it has no outlet on record). None was confirmed exactly as first written. Of 6 proposed "do not" warnings, 5 were confirmed and 1 was narrowed (fans are fine in humid heat; depending on the grid is not). **Biggest corrections:**

EXECUTIVE SUMMARY — AT A GLANCE



15.40°N 87.80°W, Honduras · Natural Earth

- **The risk.** The Ulúa comes through levee breaches open since 2020, breaking through again on 8 October 2026; dry-season water storage then breeds dengue, which killed at least 3 El Progreso children in 2024.
- **First action.** SIT finishes its Amapa and Naranjos Chinos contract and seals the San José del Cayo and Miller breaches, while the Alcaldía and CODEM make temporary closures, by 30 November.
- **The money.** $20,000 floor, $99,787 target, $123,617 ceiling; the top-ranked placement is aldea household and school chlorination through the juntas de agua, $25,000.
- **Do not.** Do not stand down flood preparations because El Niño means drought: the Ulúa broke the levee on 8 October under a green alert, and cold fronts still flood northern Honduras.

What to do	**14** actions	p. 4
Who does what	RACI chart	p. 16
Where money does the most	funding	p. 16
Do not do this	**6** items	p. 18
What has happened here	the record	p. 21
How this was produced	methodology	p. 23
References	sources	p. 24

* Suggested formulation: check with a health worker before sharing.

- The Ulúa levees are repaired by the national infrastructure ministry (SIT), not the municipality. Its current contract covers Campo Amapa and Naranjos Chinos. The municipality's new L1,500 million borrowing authority allows drainage and "vulnerability reduction" but names no levee, and no loan will be spendable before the peak.
- El Progreso's urban water is run by SANAA, the national utility, not a municipal company. The dry-season plan has to be asked of SANAA's regional office.
- No study shows that dengue training alone cuts deaths. It stays high on the list because El Progreso's deaths happened after referral, but its benefit is stated as plausible, not proven.
- Red bean prices usually peak in June–August, before the first harvest, not in February–April. Food plans are timed to that.
- El Progreso is not outside the drought alert system: all 11 Yoro municipalities were on yellow or green drought alert on 26 August. It is outside every pre-arranged El Niño fund, which all go to the Dry Corridor.

The funding target is **$99,787** (floor $20,000, ceiling $123,617), down 19% from the $123,617 first proposed, inside the one-third flag. Three of its four rows are allocations, not costings. **The El Niño premise was tested, not assumed,** from 27 dry seasons of records at San Pedro Sula airport. **Scope:** the request named Honduras; this briefing covers El Progreso, the home town of the requesting foundation's founder, at municipal resolution. **Exposure figure:** Climate Impact Lab, September 2026, pre-peer-review, publishes none for Honduras, and heat is not the main killer here.

DOES EL NIÑO MATTER HERE?

THE PREMISE, TESTED

We pulled the daily record for La Mesa, San Pedro Sula's airport, about 15 km west of El Progreso and on the same valley floor (NOAA GHCN-Daily, 27 usable dry seasons, 1984–2025). Seasons run December to May and are classed by the strength of the El Niño that peaked the winter before.

* Suggested formulation: check with a health worker before sharing.

DRY SEASON AFTER…	SEASONS	DEC–MAY RAIN	MAR–MAY MEAN TEMPERATURE	SHARE OF MAR–MAY DAYS ≥ 38 °C
Strong El Niño (2009–10, 2015–16, 2023–24)	3	230 mm	28.5 °C	33%
Moderate or weak El Niño	5	204 mm	28.4 °C	12%
Neutral	8	335 mm	28.0 °C	16%
La Niña	11	395 mm	27.7 °C	8%
All seasons	27	323 mm	28.0 °C	13%

The direction is clear, the size is not. After El Niño winters the valley's dry season has been about 30% drier and half a degree hotter, with more days at 38 °C. Three strong seasons is a small sample, and the 2024 heat drives much of the hot-day figure. The station logs maximum temperature on only about 130 days a year, so hot-day shares are of observed days. The 1982–83 and 1997–98 El Niños fall in gaps in the record.

The local record agrees. In September 2015, SANAA reported an 87% drop in flows at the city's surface sources, "especially the Río Pelo", El Progreso's main source, and named El Progreso the worst-affected city; it blamed El Niño (La Prensa, 3 September 2015). Nationally, El Niño's drought falls hardest on the south and centre. In the north, rain falls short but less severely.

EXPOSURE, HONESTLY

WHO IS AT RISK, AND WHERE

- **People behind the levees, now.** In June 2025 residents counted 16 breaches in the old banana-camp levees (La Duarte, Campo Amapa, Naranjo Chino, Miraflores, Las Cuchillas, San Isidro). The minister said SIT had no funds. On 8 October 2026 the Ulúa rose from 2.0 to 3.8 m at the Puente de la Democracia gauge. The levee broke at San José del Cayo and Miller, and water seeped into colonia Policarpo Paz García, while COPECO's alert stayed green. In low colonias residents chip in L100 each for pump fuel.
- **Children, from dengue.** In 2019, Honduras's worst epidemic, El Progreso had 5 of Yoro's 14 dengue deaths, and 59 of the country's 89 laboratory-confirmed deaths were children under 15. In July 2024 a 7-year-old girl referred from El Progreso arrived in shock at the San Pedro Sula hospital and died. An 8-year-old from El Progreso and a 9-year-old from aldea El Castaño also died. Nationally, children aged 5–9 have twice the odds of severe dengue that people over 60 do (Zúniga-Gutiérrez et al., 2024).

* Suggested formulation: check with a health worker before sharing.

- **Households on stored water.** SANAA serves about 12,000 accounts here: about 4,000 from the El Salto plant on the Río Pelo, the rest from 39 pumping stations. Thirty aldeas were reported drinking contaminated water. In urban Brazil, dengue rose 3–5 months after drought (peak relative risk 1.43; Lowe et al., 2021), because people store water in open containers.
- **The old banana camps, doubly.** In March 2026, armed groups of up to 30 men were stopping vehicles from Aldea 45 to Urraco, including La Fragua and Birichiche; charity workers were robbed, and neighbourhood council (patronato) members resigned. These are the same camps behind the open levees.
- **Outdoor workers in the March–May heat.** San Pedro Sula reached 39 °C on 27 May 2024. Cane, banana, and palm work is heavy, but no heat-death or kidney-disease data were found for El Progreso's workers.

Money: households carry the risk. Yoro holds 7.3% of the country's remittance recipients, averaging US$413.5 a month (Central Bank survey, August 2026), and remittances were 34.3% of GDP in the second quarter. In July a decree cut the electricity subsidy from 987,263 households to 318,838, just before the hot season. No farm or heat insurance reaches Yoro.

HOW FOOD REACHES EL PROGRESO

SUPPLY CHAIN

El Progreso buys its food. It is a valley market town of plantations, not a Dry Corridor grain producer, so El Niño reaches it as prices. In August white maize was 15.4% above a year earlier and about 39% above its five-year average; red beans were 18.5% above average. Diesel was up 51.5% and transport inflation was 15.8%, which raises the cost of every sack moving up from Puerto Cortés and San Pedro Sula (FEWS NET, September 2026). The second (postrera) bean and maize crop was planted late and will likely yield below average.

Beans are usually cheapest in December–February, when the postrera crop reaches market, and dearest in June–August, before the first harvest. The national school-meal programme reaches 1.2 million children but stops for the long school holiday, which normally runs from late November to early February. FEWS NET projects areas outside the Dry Corridor at IPC Phase 2 (stressed) through January 2027. Markets are working, so cash or vouchers beat food parcels here.

WHAT THIS PLACE SHOULD DO

RANKED BY LIVES PER EFFORT

Ranked by expected lives saved per unit of effort before the peak. Backing lenses are shown per action. Several of these name the Foundation for Education in Honduras (FEIH), which asked for this briefing and builds schools in and around El Progreso. Its school committees are a

* Suggested formulation: check with a health worker before sharing.

delivery channel that already has families' trust.

1. Close the open Ulúa levee breaches before the cold-front rains: SIT finishes its Amapa and Naranjos Chinos contract and seals San José del Cayo and Miller; the municipality makes temporary closures in the meantime

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OWNER	BY WHEN	COST	PREVENTS
SIT (works contract); Alcaldía and CODEM (temporary closures)	30 November	SIT's Stage III contract is L49.7 million (already let); temporary closures not quoted	The next rise of the Ulúa flooding the old banana camps through known gaps

The breaches are documented, and five days before this briefing the river broke through at San José del Cayo and Miller. SIT holds the repair contract (L49.7 million to Amacon, 150 days, covering Amapa and Naranjos Chinos); only 6 of 32 valley levee projects were finished by mid-June 2026. Temporary earth-and-sack closures by municipal machinery, coordinated with SIT under emergency powers, are feasible within weeks. The municipality's L1,500 million borrowing authority (Decreto 64-2026) permits storm drainage and "vulnerability reduction", so ask for part of it to be ring-fenced for drainage. But the lead projects are a boulevard and a commercial district, and the money will not arrive before February.

* Suggested formulation: check with a health worker before sharing.

Why it ranks first in an El Niño briefing: El Niño does not stop the cold fronts. During the 2015 El Niño, forecasters said fronts would be "fewer, but intense" in rain on the Atlantic coast. Cold-front floods in November 2025 affected about 17,900 people, with Yoro on yellow alert. And the historian's point holds: the dry season is the only window to repair before the 2027 rains. Whether El Niño makes winter fronts wetter or drier here is unresolved.

2. Get Hospital El Progreso and the aldea health centres ready for severe dengue before the rains: the PAHO warning-sign triage and fluid protocol, supplies, and monitoring beds, so children are stabilised here instead of referred in shock

PARTIALLY CONFIRMED | PUBLIC HEALTH · PHILANTHROPIST · FRONTLINE | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
SESAL Región Sanitaria de Yoro; Hospital El Progreso	Training by March 2027; supplies before May	$30,000 allocation for supplies (not a costing); the PAHO course is free	Child deaths from dengue shock during the 2027 season

The deaths are local and they happened after referral. The PAHO 2009 warning-sign guideline is the standard; a Honduran SESAL–PAHO severe-dengue course trained 72 staff in October 2025, and the PAHO online course is free. The often-quoted fall in severe-dengue case fatality "from about 10% to under 1%" with good care is a statement cited in a Puerto Rico review (Tomashek et al., 2012), not a measured trial result. No controlled study shows that training alone cuts deaths; PAHO's estimate of about 3,300 deaths avoided across the Americas in 2011–14 is ecological.*

* Suggested formulation: check with a health worker before sharing.

Transferability: the guideline was built for exactly this setting. The risk is the hospital, not the protocol: it has 122 beds for a catchment of about 350,000 across four municipalities, with a 2021 prosecutor's raid over undelivered equipment and a 2023 occupation that removed its director. Supplies and monitoring beds matter as much as training. Dengue rises three to five months into drought and peaks in the June–October rains, so this is a before-May job.

3. Keep every stored-water container covered and scrubbed weekly, from December, and pair any water handed out with a lid: through parishes now and FEIH school committees once term starts in February

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OWNER	BY WHEN	COST	PREVENTS
SESAL Yoro vector programme; parishes; FEIH school committees	December to June	Volunteer time; lids bought only alongside water distribution	The drought-to-dengue rise of March–June 2027

The Camino Verde cluster trial in Managua and Guerrero (BMJ 2015) cut dengue infection in children aged 3–9 by 29.5% (95% CI 3.8–55.3) and the house mosquito index by 44.1%, through community container work without chemicals. Covering containers was associated with far fewer larvae in Mexico (odds ratio 0.22), and WHO's standing advice is to cover, empty, and clean water containers weekly.

* Suggested formulation: check with a health worker before sharing.

The catch is cost. Camino Verde cost about US$29,000 per healthy life-year (DALY) in both settings, rated not cost-effective in Nicaragua (Tschampl et al., 2020). That is why this is volunteer-run and attached to water distribution and school routines, not funded as a stand-alone campaign. A school or parish version has never been trialled, so the 29.5% does not carry over automatically. Schools are on holiday from late November to early February, so December starts with parishes.

4. Get a dry-season water plan from SANAA's regional office by January: a published rationing schedule, tankers arranged in advance, and chlorination through the aldeas' water boards

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OWNER	BY WHEN	COST	PREVENTS
SANAA (noroccidente region); juntas de agua in the aldeas	31 January	$25,000 allocation for chlorination (recipient still to be named)	A repeat of 2015's unplanned shortage; diarrhoea from untreated water

The 2015 precedent is direct: same city, same river, an El Niño year, 87% of flow gone. El Progreso is one of the few systems SANAA never transferred to the municipality, so the plan has to come from SANAA, and no 2026 schedule or tanker contract was found. In the aldeas, more than 120 juntas de agua ran Yoro's rural systems in 2013. A meta-analysis of 18 trials with mortality data found household water treatment cut under-five deaths by about a quarter (odds ratio 0.76), at about US$27–66 per DALY (Kremer et al.); those trials ran mostly in Africa and South Asia.

* Suggested formulation: check with a health worker before sharing.

Feasibility: asking costs nothing. The chlorination money needs a named recipient, a federation of water boards or the municipality, before release; FEIH does not do water work. The headwaters matter too: by 2020, 15 of 28 water sources in the Mico Quemado reserve had dried up after fires and logging, and the municipality ran about 40 firefighters and rangers in that dry season. Those patrols are a February–May job.

5. Wire each school into CODEM's flood warning, check every designated shelter against the Eta and Iota flood footprint, and keep a paper list of who needs help to leave

PARTIALLY CONFIRMED · HISTORIAN · FRONTLINE · EMERGENCY MGR · SKEPTIC · ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Alcaldía and CODEM; FEIH and school committees	30 November	Near zero	Late evacuation, and sheltering families in a school that floods

COPECO says Ulúa alerts now give about 18–19 hours of warning from upstream stations in Intibucá and reach El Progreso's local emergency committee. Each school needs a named key-holder, drinking water, and a paper list per catchment of older people living alone, bedridden people, and children living with grandparents while parents work abroad. Paper, because phones are searched at checkpoints in the camps.

* Suggested formulation: check with a health worker before sharing.

Two cautions from the record. On 8 October the levee broke while the gauge read green, so a green alert is not a reason to wait. And El Progreso shelters people in schools (92 shelters in November 2020), but no published list says which schools sit inside the flood footprint. Get CODEM's list and check it. During Iota about 20,000 people in nearby Villanueva refused to evacuate.

6. Bridge the school-holiday meal gap with pulpería vouchers or cash for pupils' families, indexed to bean prices, from late November to early February

PARTIALLY CONFIRMED | FOOD SUPPLY CHAIN | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
FEIH and school committees	Lists by mid-November	About L550–600 per family per 25 lb of beans at market prices; pupil count not yet set, so unfunded	Children eating last in the months when school meals stop

The national programme feeds 1.2 million children in term and stops in the holiday. Even in term, an audit found meals served on only 143 of 170 school days in 2024. Markets are working (IPC Phase 2), so vouchers spent at local shops beat parcels and keep money in the neighbourhood. WFP ran take-home school rations through school committees in 2020–21, so the channel has been used.

Index the voucher to the San Pedro Sula wholesale bean price (SIMPAH, daily), because beans and maize are running hot. Target on published criteria, not on party or bono lists (see "Do not do this").

7. Pre-position oral rehydration salts and zinc at every health centre before December

PARTIALLY CONFIRMED | PUBLIC HEALTH | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
SESAL Región Sanitaria de Yoro	30 November	Low; inside existing supply lines	Diarrhoea deaths in under-fives during water rationing

Observed studies found oral rehydration cut diarrhoea deaths by 69% (95% CI 51–80%); the widely quoted 93% is a modelled ceiling (Munos et al., 2010). Zinc cuts diarrhoea deaths by an estimated further 23% (Fischer Walker and Black, 2010).*

* Suggested formulation: check with a health worker before sharing.

The evidence is from children under five in low- and middle-income countries. School-age children carry much lower risk, so the stock belongs at health centres and with families of young children, not mainly in schools.

8. Fund Radio Progreso and ERIC-SJ to broadcast dengue warning signs, water safety, and heat advice, and to deliver household kits, priced at 2026 costs

PARTIALLY CONFIRMED | FRONTLINE · PHILANTHROPIST | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
ERIC-SJ / Radio Progreso	Grant signed by December; released on a rationing or price trigger	$29,787 at twice its proven scale (2020 prices; re-quote)	Late care-seeking for dengue; families missing warnings official channels never reach

The Jesuit radio station, founded in El Progreso in 1956, is the voice people here trust. In 2020 a $20,000 Development and Peace grant paid for 500 household kits at $23.83 and 20 radio programmes at $297.84, reaching an audience put at 500,000. The message that saves lives: abdominal pain, bleeding, or persistent vomiting with fever means go to the clinic now.*

The 2020 kits included food and hygiene items (with chlorine) for hurricane victims; a 2026 kit needs a fresh quote. Its only recorded grant is $20,000, so scaling beyond twice that is unproven. No 2026 news on the organisation was found; confirm directly that it can receive the funds.

9. Plan school heat days now: the shortened 7–11 am timetable Educación used in April–May 2026, drinking water and shade at every school, and no plan that depends on the power staying on

PARTIALLY CONFIRMED | SKEPTIC · PHILANTHROPIST | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
SEDUC (Educación); FEIH for its schools	Before March	$15,000 allocation for water storage and shade at FEIH schools	Heat strain in children; lost school days

* Suggested formulation: check with a health worker before sharing.

On 28 April 2026 Educación cut the school day to 7–11 am in 17 of 18 departments and moved physical education indoors; in mid-May some classes moved online. It has the authority and has used it. Heat measurably harms learning (about 1% of a year's learning per 1 °F hotter school year without cooling, Park et al., 2020).

No study shows that shortened timetables, shade, or passive design reduce heat illness in children, so this ranks below the water and dengue work. Passive retrofits will not be finished before March–May 2027; water and shade can be.

10. Run a help desk for families dropped from the electricity subsidy, once ENEE's El Progreso office confirms how the review works

PARTIALLY CONFIRMED | ECONOMIST | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
ENEE (review); FEIH school committees (help desk)	Ask ENEE in November; desk open before the hot season	Volunteer time	Families switching off fans and fridges in the heat to save money

Decreto Ejecutivo 020-2026 cut the 150 kWh subsidy to 318,838 households, removing about 668,000. An earlier decree (12-2026) ordered ENEE to open a review mechanism, with the subsidy paid 60% by the finance ministry and 40% by ENEE.

No procedure, documents, or deadline has been published, so the first step is a visit to the ENEE commercial office. The value per household is roughly a month of bills: worth having, not resilience.

11. Make three calls: the Honduran Red Cross about its Sula valley flood protocol, COPECO for El Progreso's exact drought alert level, and WFP and SAG about adding Yoro to the 2027 farm microinsurance

PARTIALLY CONFIRMED | ECONOMIST | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
COPECO and the Honduran Red Cross; FEIH makes the calls	October	Phone calls	Missing pre-arranged help that already covers this valley

* Suggested formulation: check with a health worker before sharing.

No pre-arranged El Niño money names El Progreso: the UN's CERF drought framework (US$4 million) covers five municipalities in El Paraíso and Francisco Morazán, and the government's US$70 million fund and the UN's US$63.2 million plan target the Dry Corridor. The Red Cross drought protocol (EAP2023HN02) picks its municipalities at activation and did not activate in Honduras in 2026. The better fit is its flood protocol EAP2025HN05 (CHF 917,330, 10,000 people), which covers the Ulúa basin and Sula valley and releases money 3–5 days before a forecast tropical storm.

The SAG/WFP crop microinsurance covers seven departments, not Yoro; if Yoro is added, push for its excess-rain leg, since flood is the local peril. The CCRIF payout after Storm Sara (US$4.67 million) went to the treasury, not to households.

12. Buy beans for school feeding at the December–February low and store them in hermetic bags; help hill-aldea bean growers get bags before the harvest

PARTIALLY CONFIRMED | FOOD SUPPLY CHAIN | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
FEIH, coordinating with SEDESOL and WFP	**Bags by November; buying December–February**	**Not quoted; get a SIMPAH wholesale price and a bag price first**	**School feeding priced out in the June–August peak; storage losses**

Beans reach market from the postrera crop in December–February and peak in June–August. With late planting this year the low may be shallow and late. A Zamorano study found Honduran smallholders lost 8.5% of stored beans by weight over seven months; hermetic storage holds maize losses to about 2%, though bean evidence is thinner.

* Suggested formulation: check with a health worker before sharing.

The state grain agency lost about 44,000 quintales of beans to spoilage, which is the warning: buy only with proper storage, and top up the national meal programme rather than duplicate it. The valley floor grows banana, sugar, and palm; beans come from the hill aldeas, so map them first.

13. Ask cane mills, banana farms, and palm estates to adopt water, rest, and shade for field crews from March to May

PARTIALLY CONFIRMED | PUBLIC HEALTH | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Employers, with La Isla Network	Approach by January; run March–May	Employer-borne	Heat strain and acute kidney injury in field workers

At one Nicaraguan mill, burned-cane cutters saw incident kidney injury fall from 27% to 7% (RR 0.3) after a water-rest-shade programme (Glaser et al., 2020). Seed cutters saw no benefit, where it was patchily run.

The study was a before-and-after comparison without a control, and nothing has been evaluated for banana or palm work. Honduras has no binding heat standard; certification schemes give buyers leverage. La Isla Network launched Latin American worker-heat programmes in 2026.

14. Ask the state grain agency and BANASUPRO to extend discounted bean and maize sales to El Progreso

UNVERIFIED | FOOD SUPPLY CHAIN | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
IHMA and BANASUPRO; Alcaldía makes the request	October request	None to the requester	Price exposure for the poorest buyers

IHMA and BANASUPRO were negotiating a 25% discount at sale points in August. BANASUPRO's Sula valley outlets and mobile routes are in San Pedro Sula; no outlet or sale in El Progreso is on record.

* Suggested formulation: check with a health worker before sharing.

The agency's delivery record is weak (spoiled stocks, unpaid producers, two investigations). Make the request because it is free, but do not plan around it.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	SIT	Alcaldía & CODEM	COPECO & Red Cross	SESAL & Hospital	SANAA & juntas de agua	FEIH & school committees	ERIC-SJ, Radio Progreso & parishes	SEDUC	ENEE	Employers	IHMA, SEDESOL, WFP & SAG
1. Close levee breaches	**A/R**	**R**	**C**			**I**	**I**				
2. Dengue readiness		**I**		**A/R**		**C**	**I**				
3. Covered water containers				**A**	**C**	**R**	**R**				
4. Dry-season water plan		**C**		**C**	**A/R**	**C**					
5. Schools in the flood warning		**A**	**C**			**R**		**C**			
6. Holiday meal vouchers						**A/R**	**I**				**C**
7. ORS and zinc				**A/R**		**I**					
8. Radio and kits			**C**	**C**		**I**	**A/R**				
9. School heat days				**C**		**R**		**A**			
10. Subsidy help desk						**R**	**C**		**A**		
11. Three calls		**C**	**A**			**R**					**C**
12. Beans and storage						**A/R**					**C**
13. Water, rest, shade				**C**						**A/R**	
14. Discount grain sales		**R**				**C**					**A**

What the chart shows: FEIH and its school committees carry the most rows (nine with work or sign-off), which suits a foundation whose schools families already trust. SESAL is accountable for the three health rows that save the most lives. The top row needs a decision and a contract from SIT, not donor money, and rows 4, 10, and 11 need only a request.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs.

FLOOR

$20,000: one channel at the one scale it has proven

TARGET

$99,787

CEILING

$123,617

LARGEST UNCERTAINTY

Whether the hospital and a water-board federation can take and spend the money; neither has a record of doing so

* Suggested formulation: check with a health worker before sharing.

Target: ERIC-SJ at twice its 2020 scale (1,000 kits × $23.83 = $23,830, plus 20 programmes × $297.84 = $5,957, for $29,787), plus $25,000 chlorination, $30,000 hospital dengue readiness, and $15,000 school water and shade = $99,787. Ceiling: the same with ERIC at four times its 2020 scale ($53,617) = $123,617; above that, no channel has shown capacity. The target fell 19% from the $123,617 first proposed, inside the one-third flag. Three rows are allocations, not costings: they are judgements of what each channel could absorb, to be priced before release. The holiday meal vouchers (action 6) are not in the envelope because the number of pupils is not yet set. No placement duplicates a pre-arranged fund: CERF, the government drought fund, and the UN plan all cover other municipalities.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
A federation of El Progreso's juntas de agua, or the municipality (to be named)	Action 4: household and school chlorination in the aldeas	$25,000 (allocation)	Grant released when dry-season rationing begins	Water treatment: under-five deaths down about a quarter, US$27–66 per DALY (Kremer et al., meta-analysis of trials mostly in Africa and South Asia)	PARTIALLY CONFIRMED
Hospital El Progreso, via SESAL Yoro and a fiscal sponsor	Action 2: dengue triage supplies, IV fluids, monitoring beds	$30,000 (allocation)	In-kind grant	No published cost per life; El Progreso's child deaths followed referral in shock. The hospital's support group has received donations before, which is a lead, not a channel	PARTIALLY CONFIRMED
ERIC-SJ / Radio Progreso	Action 8: warning broadcasts and household kits	$29,787	Pre-signed grant, released on a rationing or price trigger	2020 unit costs from its $20,000 Development and Peace grant; re-quote at 2026 prices	PARTIALLY CONFIRMED
FEIH	Action 9: water storage and shade at its schools	$15,000 (allocation)	Direct spend	No unit cost or health evidence; FEIH spent $179k in 2024, so this is within its scale	PARTIALLY CONFIRMED

The funder's one thing: El Progreso sits in a hole in the map. Every pre-arranged El Niño dollar in Honduras goes to the Dry Corridor and the west, so money here adds rather than duplicates. That only holds if it is signed now and released on a trigger. And do not pay for neighbourhood mosquito campaigns: the best-evidenced one cost about US$29,000 per DALY, hundreds of times more than water treatment.

* Suggested formulation: check with a health worker before sharing.

Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for this hazard and would cost lives, or waste the window, here.

"El Niño means drought, so stand down flood preparations"

CONTRAINDICATED

The Ulúa broke through the levee on 8 October 2026 while COPECO's alert read green, and the 16 openings counted in June 2025 have not all been sealed. Cold fronts still flood northern Honduras in El Niño winters; November 2025's affected about 17,900 people. Mitch struck in 1998, in the La Niña that followed El Niño, which is a warning for the 2027 transition season.

Instead: keep flood readiness through the November–February cold-front season, treat a green gauge as compatible with a local breach, seal the known openings now, and check every designated shelter against the Eta and Iota flood footprint.

Handing out open barrels or uncovered tanks for dry-season water storage

CONTRAINDICATED

Barrels and tanks produce most of the mosquitoes in Central American and Mexican surveys (74% of pupae in one Mexican study), and poorly fitting lids still breed them. Drought-era storage is how drought becomes a dengue epidemic three to five months later.

* Suggested formulation: check with a health worker before sharing.

Instead: distribute only containers with tight-fitting lids or mesh covers, preferably sealed tanks with taps, with a weekly empty-and-scrub routine.

Buying or lobbying for insecticide fogging trucks to stop dengue

CONTRAINDICATED

In four Tegucigalpa neighbourhoods, permethrin killed only 1–48% of Aedes aegypti, malathion 24–74%, and the main resistance gene was fixed in every population (Parasites & Vectors, 2025). A systematic review found no clear evidence that space spraying works as a single intervention, and no randomised trial of fogging against dengue exists. No Sula valley resistance data exist either, which is the point: nobody knows the trucks work here.

Instead: pay for container work and larval control in stored water, and test local mosquitoes for resistance before buying any adulticide. If fogging is used in an outbreak, move away from permethrin and malathion.

Choosing who gets aid from bono lists or patronato leaders alone

CONTRAINDICATED

The Bono de Alivio Climático paid L661.1 million to 101,453 households (19,909 in Yoro) during the 2025 campaign; recipients reported being told how to vote, and beneficiary lists were never published (allegations, under no formal finding). In the old banana camps, patronato members have been forced out by armed groups, so a patronato list there may reflect who the gang allows.

* Suggested formulation: check with a health worker before sharing.

Instead: target on published, checkable criteria such as flood-zone address, COPECO's damage census, school rolls, and health-centre registers. Publish the list and add a complaint channel outside party structures.

Sending outside volunteers door to door through the Aldea 45 to Urraco camps

CONTRAINDICATED

In March 2026, groups of up to 30 armed men were stopping and searching vehicles from Aldea 45 to Urraco, with La Fragua and Birichiche at the centre; members of charitable organisations were robbed. This applies to that corridor; it is not shown for the rest of the municipality.

Instead: deliver through fixed places people already use (the health centre, the church, the school) and through people they know, such as the teacher or a grandmother. Run a security assessment first, carry no visible cash or goods, and do not ask patronato leaders to front the work.

Heat plans that rely on air-conditioned centres or cold storage staying powered (narrowed)

CONTRAINDICATED

In May 2024 heat and drought cut night-time generating capacity to about 1,750 MW, with outages for weeks, and the July 2026 subsidy cut makes running power dearer for poor households. Fans themselves are not the problem: in humid heat like El Progreso's they help, and they become harmful only at very high dry-bulb temperatures, around 38–39 °C.

Instead: plan for outages. Use shaded, ventilated spaces, water and wetting the skin, check-ins on older people, and backup power at clinics and any designated cooling site. Fans are fine; just not as the only measure.*

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
1974	Hurricane Fifi; El Progreso and Choloma among the hardest-hit places in the Sula valley	More than 8,000 nationally; no municipal count found	Not established
1998	Hurricane Mitch, in the La Niña that followed the 1997–98 El Niño. The Ulúa overtopped completely, the Pelo destroyed homes in barrio Pénjamo, and Urraco was cut off without safe water	5,657 to more than 7,000 nationally, by source; no municipal count found	El Tablón dam studies promised since 1997; not built
2015	El Niño water shortage: SANAA reported an 87% drop in surface-source flows, especially the Río Pelo; El Progreso the worst-affected city	None recorded	Nothing documented for El Progreso
2019	Dengue epidemic, Honduras's worst (about 180 deaths)	5 deaths in El Progreso, the most in Yoro; nationally 59 of 89 lab-confirmed deaths were children under 15	Nothing documented locally
2020	Eta and Iota. The Ulúa flooded the old banana camps (Campo Buena Vista, Río Chiquito with 33 houses destroyed, Amapa, Naranjo Chino); 92 shelters opened. Separately, 15 of 28 Mico Quemado water sources had dried up by February	95 nationally; no municipal count found	Levee breaches left open; 16 counted in June 2025
2023	November cold-front floods; CODEM opened 16 shelters for about 1,500 people	None recorded	Residents in low colonias still pool money for pump fuel
2024	Dengue (July–August): a 7-year-old arrived in shock in San Pedro Sula and died, and an 8-year-old and a 9-year-old (El Castaño) died. Tropical Storm Sara (November): the Amapa levee gave way	At least 3 children from dengue	CCRIF paid the government US$4.67 million for Sara; nothing reached the levees

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
2026	8 October: Ulúa from 2.0 to 3.8 m; levee broke at San José del Cayo and Miller with the alert at green	None reported	To be decided this winter

The recurrence nobody has noticed: in El Progreso, the El Niño year is not the deadly one; the year after is. The river shrinks in the El Niño dry season, everyone stops worrying about it, the levees stay broken, and the next wet season comes through them. Mitch arrived in the La Niña after the 1997–98 El Niño. The drought of winter 2026–27 is the repair window for the floods of 2027.

WHERE THE EXPERTS DISAGREE — RESOLVED AND OPEN

- **Fund community mosquito-container campaigns** (public health, skeptic: a 29.5% cut in child infection in a trial) versus **do not fund them** (philanthropist: about $29,000 per DALY). Resolved here by running the work through volunteers, schools, and any water distribution, without stand-alone funding.
- **The hazard is drought and heat** (the original framing; the station record shows drier, hotter dry seasons after El Niño) versus **the hazard is flood, now** (emergency manager, skeptic, historian). Both hold, in sequence: flood through the cold-front season, water shortage from December, dengue from March. Open: whether El Niño makes winter fronts wetter or drier here; one 2015 forecast said "fewer, but intense", and no study settles it.
- **Stock schools as shelters** (emergency manager) versus **do not shelter people behind the levees** (skeptic). Resolved by checking CODEM's shelter list against the Eta and Iota flood footprint first; no published list places any school in or out of it.
- **Heat will kill this winter** versus **the deaths will be dengue, in mid-2027** (public health). The record favours dengue: no heat death has been documented in El Progreso, while children died of dengue in 2019 and 2024. A heat-only mortality model would miss them.

WHAT NOBODY COVERED — BLIND SPOT

No lens examined the maquila factories and their workers: how hot the plants get, whether they keep running in outages, and whether the factory floor could carry dengue warnings and water messages to thousands of workers at once. Nor did any lens test the town's real insurer, its emigrants. Remittances are uncorrelated with drought but directly exposed to US deportation and labour policy, and a fall in them during a high-price year would hit El Progreso harder than any rainfall shortfall. Either could change who is most at risk.

* Suggested formulation: check with a health worker before sharing.

HOW TO READ THIS

METHOD

The panel is author-built. Eight simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are resolved at Climate Impact Lab impact-region scale, roughly 300,000 people. They are not neighbourhood figures. The El Niño projection covers *heat only*; drought, flood, fire, smoke, and crop failure are additional to it and are carried here by the historical record rather than by a published number.

HOW THIS BRIEFING WAS PRODUCED

METHODOLOGY

- **Scope.** The request named Honduras, a whole country. The briefing was built for El Progreso, Yoro, the home town of the requesting foundation's founder, at municipal resolution: the town, its colonias, and its aldeas. Climate Impact Lab impact region, as the report records it: the region containing El Progreso, Yoro (not resolved by name). Mortality figures exist only at that scale (impact regions average roughly 300,000 people); sub-municipal detail is used for hazard exposure and action design, not for death counts.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Where heat is not the dominant hazard, the historical record carries the hazard instead of a borrowed number.
- **Premise.** Dry-season rain and temperature after each class of ENSO winter were computed from NOAA GHCN-Daily station HOM00078708 (La Mesa, San Pedro Sula), classing winters by the October–February peak of the Oceanic Niño Index (strong at 1.5 or above).
- **Expert panel.** Eight simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, philanthropist, and food supply chain analyst) each researched this place independently and in parallel, from web sources in Spanish and English, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.

* Suggested formulation: check with a health worker before sharing.

- **Verification.** Seven verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist. Food actions were checked for price-series frequency, peak timing, channel capacity, and cash versus in-kind. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed.
- **Who does what.** The RACI chart is built from the owners named in the verified actions, with one accountable party per action. It is a proposal: no organisation named in it has been contacted or has agreed to a role.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Not consulted. The inventory loads its entries in the browser, and every fetch from this run returned an empty shell ("Number of Adaptations: 0"); the downloadable dataset was not retrieved. The tags therefore say "not consulted", which is not the same as "no precedent".
- **Health wording.** Passages where a wording error could hurt someone (dengue warning signs and when to seek care, oral rehydration salts and zinc, and cooling during power outages) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. La Tribuna returned errors on direct fetches (its claims were checked through search results and other outlets), and the SIMPAH price site did not resolve. COPECO's municipal drought-alert table, CODEM's shelter list, and El Progreso's 2026 municipal budget were not found. No municipal death toll was found for Fifi, Mitch, or Eta and Iota; a reported figure of 254,000 people unable to return home after Eta and Iota exceeds the municipality's population and was discarded.

References · Premise: NOAA GHCN-Daily, La Mesa · CPC Oceanic Niño Index [computed] · Floods: La Prensa, levee breaches, 18 Jun 2025 [confirmed] · La Prensa, Ulúa rise, 8 Oct 2026 [confirmed] · La Tribuna, 8 Oct 2026 [not fetched, 403] · La Prensa, SIT levee projects [confirmed] · La Gaceta 37,145, Decreto 64-2026 [confirmed] · La Prensa, Ulúa warning time [confirmed] · La Prensa, Villanueva evacuation [confirmed] · Contracorriente, Policarpo Paz [corrected] · La Prensa, Iota shelters [confirmed] · El Heraldo, Eta and Iota [partially; population figure discarded] · La Prensa,

* Suggested formulation: check with a health worker before sharing.

cold fronts 2015 · La Prensa, cold-front floods Nov 2025 · IRI, Mitch and ENSO · PC(USA) History, Fifi · Envío, Mitch in El Progreso · El Heraldo, El Tablón [unverified] · Water: La Prensa, 3 Sep 2015 [confirmed] · La Prensa, Mico Quemado, 25 Feb 2020 [corrected] · La Prensa, SANAA El Progreso · La Prensa, SANAA transfers · La Prensa, 30 aldeas · La Prensa, juntas de agua · Kremer et al., water treatment [confirmed] · Health: El Heraldo, dengue 2019 [confirmed] · La Prensa, 24 Jul 2024 [confirmed] · La Prensa, 29 Jul 2024 [corrected] · Zúniga-Gutiérrez et al. 2024 [confirmed] · Tomashek et al. 2012 [corrected] · PAHO, warning signs · PAHO, Honduras ICU course 2025 · BMC Med Educ 2016 · El Heraldo, hospital director · UNITEC, Hospital El Progreso [secondary] · Camino Verde, BMJ 2015 [confirmed] · Tschampl et al. 2020 [confirmed] · Lowe et al. 2021 [corrected] · Mexico container study 2017 · WHO dengue fact sheet · Munos et al. 2010 [corrected] · Fischer Walker and Black 2010 · Glaser et al. 2020 [corrected] · La Isla Network 2026 · El Heraldo, school hours [confirmed] · AFP, May 2026 · Park et al. 2020 · Contraindications: Tegucigalpa resistance, 2025 [confirmed] · Esu et al. 2010 · Bowman et al. 2016 · El Heraldo, Bono de Alivio Climático [confirmed as allegations] · La Prensa, armed groups, 12 Mar 2026 [confirmed] · E&N, outages 2024 [confirmed] · Fan modelling study, 2021 · Money: CERF anticipatory action 2026 [confirmed] · CERF framework municipalities · Government drought fund [confirmed] · TuNota, drought alerts 26 Aug · UN plan · IFRC EAP2023HN02 [confirmed] · IFRC EAP2025HN05 [confirmed] · El Heraldo, ENEE subsidy [confirmed] · La Prensa, Decreto 12-2026 · SAG/WFP microinsurance [confirmed] · CCRIF payout [confirmed] · BCH remittance survey via TuNota [confirmed] · Development and Peace, ERIC 2020 [corrected] · ProPublica, FEIH · FEIH · Food: FEWS NET, September 2026 [confirmed] · FEWS NET, bean seasonality · El Heraldo, IHMA stocks [corrected] · School meals [confirmed] · ASJ audit via La Prensa · BANASUPRO mobile units · Zamorano, bean storage losses · Hermetic storage review

Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory

Generated 13 October 2026 for El Progreso, Yoro, Honduras. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

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